Work, Energy, and Simple Machines

Important Questions with Answers

Class 9 Science | NCERT Chapter 7 | 42 Questions

MCQ · True/False · Fill in the blanks · Short answer · Numericals · Long answer

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Tip: cover the green answer box, try the question first, then check. Compare with your own textbook wording.

A. Multiple Choice Questions

Q1 - Q5

Q1. Work done on an object is zero when:

(a) force and displacement are in the same direction (b) a person pushes a rigid wall (c) a bag is lifted upward (d) a ball is kicked forward

Ans: (b) Pushing a rigid wall. There is no displacement (s = 0), so W = F × s = 0.

Q2. 1 joule is equal to:

(a) 1 N ÷ 1 m (b) 1 N × 1 m (c) 1 kg × 1 m s^{-1} (d) 1 N × 1 s

Ans: (b) 1 N × 1 m. From W = F × s, 1 J = 1 N × 1 m = 1 kg m^2 s^{-2}.

Q3. If the speed of a vehicle doubles, its kinetic energy becomes:

(a) 2 times (b) 4 times (c) 8 times (d) half

Ans: (b) 4 times. K = ½mv^2, so replacing v by 2v gives 4 × ½mv^2.

Q4. The potential energy of a 2 kg object at a height of 5 m (g = 10 m s^{-2}) is:

(a) 10 J (b) 50 J (c) 100 J (d) 200 J

Ans: (c) 100 J. U = mgh = 2 × 10 × 5 = 100 J.

Q5. The mechanical advantage of a fixed pulley is:

(a) 0 (b) 1 (c) 2 (d) more than 2

Ans: (b) 1. Effort equals load; a fixed pulley only changes the direction of the effort.

A. Multiple Choice Questions (contd.)

Q6. A goalkeeper stops a football. The work done by the goalkeeper on the ball is:

(a) positive (b) negative (c) zero (d) infinite

Ans: (b) Negative. The force applied is opposite to the displacement of the ball.

Q7. 1 watt is equal to:

(a) 1 J s (b) 1 J s^{-1} (c) 1 N m (d) 1 kg m s^{-1}

Ans: (b) 1 J s^{-1}. Power = work ÷ time, so 1 W = 1 joule of work per second.

Q8. A lever has an effort arm of 60 cm and a load arm of 20 cm. Its mechanical advantage is:

(a) 0.33 (b) 3 (c) 40 (d) 80

Ans: (b) 3. MA = effort arm ÷ load arm = 60 ÷ 20 = 3.

B. True or False

Q9 - Q14

Q9. Work is done whenever a force is applied, even if the object does not move.

Ans: False. W = F × s; if there is no displacement (s = 0), no work is done.

Q10. Work has both magnitude and direction.

Ans: False. Work has no direction; it is a number with a positive or negative sign.

Q11. The kinetic energy of a moving object can be negative.

Ans: False. $K = \frac{1}{2}mv^2$ and v^2 is never negative, so kinetic energy is zero or positive.

Q12. A stretched spring stores potential energy.

Ans: True. Work done to deform it is stored as potential energy and returned when released.

Q13. For a freely falling object (no air resistance), mechanical energy stays constant.

Ans: True. The loss in potential energy equals the gain in kinetic energy; the sum stays mgh.

Q14. A fixed pulley reduces the force needed to lift a load.

Ans: False. It only changes the direction of the force (MA = 1).

B. True or False (contd.)

Q15. A longer inclined plane (same height) needs a smaller effort.

Ans: True. MA = L ÷ h, so a larger L gives a larger MA, but the effort acts over a longer distance.

Q16. A machine can reduce the total work needed to do a task.

Ans: False. Machines reduce the force or change its direction, but never reduce total work.

Q17. 1 horsepower is equal to 746 watts.

Ans: True. 1 hp = 746 W.

Q18. A girl carrying a box while walking horizontally does work on the box by the upward force.

Ans: False. The upward force is perpendicular to the horizontal displacement, so it does zero work.

C. Fill in the Blanks

Q19. Work done = _____ × _____ (in the direction of force).

Ans: force × displacement.

Q20. The kinetic energy of a body of mass m and velocity v is _____.

Ans: ½mv².

Q21. The potential energy of an object of mass m at a small height h is _____.

Ans: mgh.

Q22. Power is the _____ at which work is done, and its SI unit is the _____.

Ans: rate; watt (W).

Q23. Mechanical advantage = _____ ÷ _____.

Ans: load ÷ effort.

Q24. The three main parts of a lever are the fulcrum, the _____ and the _____.

Ans: load and effort.

D. Short Answer Questions

Q25 - Q30

Q25. Define work. State its SI unit and define 1 joule.

Ans: Work done by a constant force = force × displacement in the direction of the force (W = F × s). The SI unit is the joule (J). 1 J is the work done when a force of 1 N displaces an object by 1 m in the direction of the force.

Q26. You push hard against a wall and feel tired. Is any work done on the wall?

Ans: No. The wall has no displacement (s = 0), so the work done on it is zero. You feel tired because your muscles use up internal energy.

Q27. State the work-energy theorem.

Ans: The work done on an object is equal to the change in its energy. Positive work increases its energy; negative work decreases it.

Q28. Differentiate between kinetic energy and potential energy.

Ans: Kinetic energy is energy due to motion ($K = \frac{1}{2}mv^2$), e.g. a moving bicycle. Potential energy is stored energy due to deformation or relative position ($U = mgh$ near Earth), e.g. a stretched spring or a ball at a height.

Q29. Why do children of different masses reach the bottom of the same slide with the same speed (ignoring friction)?

Ans: PE at the top converts to KE at the bottom: $\frac{1}{2}mv^2 = mgh$, so $v = \sqrt{(2gh)}$. The mass cancels, so speed depends only on the height h, not on the mass or the shape of the slide.

Q30. Define power. Why does running upstairs need more power than walking up?

Ans: Power is the rate of doing work, P = W ÷ t. The work done is the same, but running takes less time, so the power required is greater.

D. Short Answer Questions (contd.)

Q31. What is mechanical advantage? Why do we use a fixed pulley if its MA is 1?

Ans: MA = load ÷ effort. A fixed pulley does not reduce the force but changes its direction; pulling down is easier and more convenient than lifting up.

Q32. Why are roads on hills built to wind around in gentle slopes?

Ans: A winding road has a larger length L for the same height h, so MA = L ÷ h is larger and the force needed (e.g. by a vehicle) is smaller, though over a longer distance.

Q33. State the condition for a lever to balance.

Ans: Effort × effort arm = load × load arm ($n_1 \times L_1 = n_2 \times L_2$). Increasing the effort arm reduces the effort needed.

Q34. Why does a real pendulum slow down and stop even though mechanical energy is conserved in theory?

Ans: Friction at the support and air resistance convert some mechanical energy into heat. Mechanical energy is lost from the pendulum, though total energy is still conserved.

E. Numericals

Q35 - Q37

Q35. A force of 40 N moves an object by 5 m in the direction of the force in 10 s. Find the work done and the power.

Ans: Work $W = F \times s = 40\ \text{N} \times 5\ \text{m} = 200\ \text{J}$.
Power $P = W \div t = 200\ \text{J} \div 10\ \text{s} = 20\ \text{W}$.

Q36. A 0.2 kg ball is at a height of 10 m ($g = 10\ \text{m s}^{-2}$). Find its potential energy and its speed just before it hits the ground (ignore air resistance).

Ans: $U = mgh = 0.2 \times 10 \times 10 = 20\ \text{J}$.
At the ground all PE becomes KE: $\frac{1}{2}mv^2 = 20\ \text{J}$, so $v^2 = 200$ and $v = \sqrt{200} \approx 14.1\ \text{m s}^{-1}$.

Q37. A lever has a load arm of 0.5 m and an effort arm of 2 m. It lifts a load of 400 N. Find the effort needed and the mechanical advantage.

Ans: Effort × effort arm = load × load arm → effort × 2 = 400 × 0.5, so effort = 100 N.
MA = load ÷ effort = 400 ÷ 100 = 4 (also = 2 ÷ 0.5 = 4).

F. Long Answer Questions

Q38 - Q41

Q38. Explain positive, negative and zero work with one example of each.

Ans: • Positive work: force and displacement are in the same direction, e.g. pushing a wheelchair forward.
• Negative work: displacement is opposite to the force, e.g. a goalkeeper stopping a ball (200 N, hand moves back 0.15 m → W = −30 J).
• Zero work: when F = 0, or s = 0 (pushing a rigid wall), or force perpendicular to displacement (girl carrying a box while walking).
• Work has no direction; it only carries a + or − sign.

Q39. Derive the expression for kinetic energy, $K = \frac{1}{2}mv^2$.

Ans: Let a constant force F act on a mass m, changing velocity from u to v over displacement s. Then a is constant and $v^2 = u^2 + 2as$, so $s = (v^2 - u^2) \div 2a$.
Work $W = F \times s = ma \times (v^2 - u^2) \div 2a = \frac{1}{2}m(v^2 - u^2)$.
By the work-energy theorem this is the change in energy. For u = 0 the change equals the final kinetic energy, so $K = \frac{1}{2}mv^2$.

Q40. Show that the mechanical energy of a freely falling object is conserved.

Ans: An object of mass m is dropped from height h (point A): PE = mgh, KE = 0, so ME = mgh.
After time t (point B): v = gt and height $h' = h - \frac{1}{2}gt^2$.
$PE = mgh' = mgh - \frac{1}{2}mg^2t^2$, $KE = \frac{1}{2}mv^2 = \frac{1}{2}mg^2t^2$.
ME = PE + KE = mgh. The loss in PE equals the gain in KE, so mechanical energy stays constant.

Q41. Derive the mechanical advantage of an inclined plane. Find it for a ramp 50 cm long and 30 cm high.

Ans: Raising an object of mass m at constant speed up a ramp of length L (ignoring friction): work by effort = $F' \times L$, and PE gained = mgh. So $F' \times L = mgh$, i.e. $mg \div F' = L \div h$.
MA = load ÷ effort = $mg \div F' = L \div h$. Since L > h, MA > 1.
For the ramp: MA = 50 cm ÷ 30 cm = 1.67. A longer ramp needs less effort but over a larger distance; the work done is the same.

F. Long Answer Questions (contd.)

Q42. Name the three simple machines and explain how each makes work easier. Why can no machine give more work than is put in?

Ans: • Pulley: a fixed pulley changes the direction of effort (MA = 1); movable pulley systems give MA > 1.
• Inclined plane: a longer ramp needs a smaller force, MA = L ÷ h.
• Lever: effort × effort arm = load × load arm; MA = effort arm ÷ load arm.
In every case the force is reduced (or redirected) but the distance moved increases, so total work stays the same. Machines do not create energy; they only help us use it more conveniently.